PalmSource®
Conference and Expo



Installing Applications and Conduits

Neil Rhodes
Author and Contract Developer, Calliope Enterprises, Inc.



We are going to learn how to programmatically install applications and conduits (on Windows)



Agenda Slide

- Installing and Registering
- DLLs
- Shim DLL
- Commercial Installers
- Q&A



Installing and Registering

- The Installer may need to install:
  - Handheld databases/applications
  - And/or conduits
- For handheld databases:
  - The installer needs to provide a user list to choose from when there are multiple users
  - The installer needs to prepare databases to be installed on the next sync



Installing and Registering

- For conduits, the installer needs to:
  - Register the conduit
  - If the HotSync app is running, restart it



Uninstalling

- When uninstalling, the installer needs to:
  - Unregister the conduit
  - If the HotSync app is running, restart it
- The handheld databases will still be there
  - The uninstaller doesn't worry about them



Agenda Slide

- Installing and Registering
- DLLs
- Shim DLL
- Commercial Installers
- Q&A

DLLs—in the HotSync Folder

- Install Aide (InstAide.dll) duties:
  - Install files onto handheld
    - The install occurs on next sync
  - Get info about Palm desktop users
    - Number of
    - Names
  - Provide the path to the directory containing HotSync.exe
  - Remove from list of files to be installed

DLLs—Install Aide

- An app that uses this DLL
  - Start->Programs->Palm Desktop->Install Tool
- Common calls

```
PltGetUserCount()
PltGetUser(userIndex, nameBuffer,
  bufferSize)
PltInstallFile(userName, fileSpec)
PltRemoveInstallFile(user, fileName)
```

DLLs—Conduit Manager

- The Conduit Manger (CondMgr.dll) contains routines for:
  - Registering a conduit
  - Unregistering a conduit
  - Finding the HotSync path

DLLs—Conduit Manager

- Original Method
  - Registration was done by directly modifying the Windows Registry
  - List of keys: Application0, Application1, Application2, etc.
  - Registering a conduit
    - Created a new key at end of list
  - Unregistering a conduit
    - Removed your key
    - Renumbered all later keys

DLLs—Conduit Manager

- Original Method Problem
  - Easy to get wrong
    - One bad register/unregister could ruin the registration of all conduits
- Solution—CondMgr.DLL
  - Conduit Manager APIs interface with the Windows register
    - This could change in the future. Conduit registration information could be kept somewhere else

DLLs—Conduit Manager

- CondCfg.exe uses the CondMgr
- Common calls
  - `CmGetConduitByCreator`
  - `CmGetConduitCount`
  - `CmGetCreatorID`
  - `CmGet/SetCreator…`
  - `CmInstallCreator`
  - `CmRemoveConduitByCreatorID`
  - `CmRestoreHotSyncSettings`
  - `CmGetHotSyncExecPath`

Conduit Install Sample Code

```
#include "CondMgr.h"

int main(int argc, char ** argv)
{
  err = CmInstallCreator("Xyzz", CONDUIT_APPLICATION);
  if (err == 0)
  err = CmSetCreatorName("Xyzz",
    "C:\\MyCond\\Debug\\MyCond.DLL");
  if (err == 0)
    err = CmSetCreatorDirectory("Xyzz", "MyCond");
  if (err == 0)
    err = CmSetCreatorFile("Xyzz", "MyCond");
  if (err == 0)
    err = CmSetCreatorPriority("Xyzz", 2);
  if (err == 0)
    printf(" Registration succeeded\n");
  else
    printf(" Registration failed %d\n", err);
  return err;
}
```

Conduit Uninstall Sample Code

```
#include "CondMgr.h"

int main(int argc, char **argv)
{
  const char *kCreator = "Sles";
  int numRemoved;

  numRemoved = CmRemoveConduitByCreatorID(kCreator);
  if (numRemoved >0)
    printf("Unregistration succeeded for %d conduits\n",
      numRemoved);
  else
    printf("Unregistration failed %d\n", numRemoved);
}
```

DLLs—HotSync API

- HotSync API (HSAPI.dll)
  - **`HSGetSyncStatus`**
    - Returns idle or syncing
  - **`HSSetAppStatus`**
    - Quit
    - Start
    - Restart

DLLs—A Problem

- Where are these DLLs?
  - In Palm directory
  - How do you find the Palm directory?
    - Use Install Aide (or Conduit Manager)
    - Where is Install Aide DLL?
      - **In Palm directory**
        - **How do you find the Palm directory?**

DLLs—A Solution

- Ship with each of the three DLLs
  - Use CondMgr.DLL to find the HotSync directory
- Load the needed DLL from the HotSync directory
  - Use that DLL for your remaining calls
- If a DLL is not found, use version you shipped with
- Don’t install the DLLs into the HotSync directory!
  - A DLL in the HotSync directory matches the HotSync executable that’s there



Agenda Slide

- Installing and Registering
- DLLs
- Shim DLL
- Commercial Installers
- Q&A

Shim DLL—What Does It Do

- Provides entrypoints from:
  - HSAPI.DLL
  - CondMgr.DLL
  - InstAide.DLL
- Provides init and deinit entrypoints
  - Specifies where to find copies of the 3 DLLs
- Finds HotSync directory
- Looks for needed DLLs in that directory
  - If not found, uses copies from specified directory
- Shim DLL Source at: ftp://ftp.calliopeinc.com/palmprog2/PalmDLLShim.zip

Shim DLL (Modified Uninstall Program)

```
#include "CondMgr.h"
#include "PalmDLLShim.h"

int main(int argc, char **argv)
{
  const char *kCreator = "Sles";
  int numRemoved;

  if (!PalmDLLShimInit("C:\\PalmDLLs")) {
    fprintf(stderr, "Initialization failed\n");
    exit(1);
  }
  numRemoved = CmRemoveConduitByCreatorID(kCreator);
   if (numRemoved >0)
    printf("Unregistration succeeded for %d conduits\n",
      numRemoved);
  else
    printf("Unregistration failed %d\n", numRemoved);
  PalmDLLShimDeinit();
}
```



Agenda Slide

- Installing and Registering
- DLLs
- Shim DLL
- Commercial Installers
- Q&A



Commercial Installers

- InstallShield
- Installer Vise
- Pilot Catapult

Installers—InstallShield

- InstallShield has many flavors
  - InstallShield Professional 5.0 (no longer sold)
  - InstallShield for Visual C+
    - Free with Visual C++ 6.0. (based on IS 5.0)
  - InstallShield Express
  - InstallShield Developer 7
  - InstallShield Professional 6.3

Installers—InstallShield

- All versions of InstallShield can call a DLL at install time
- Some versions can only call functions with a specific signature:
  - InstallShield Express
  - InstallShield for Visual C++
- Other versions can call any function in a DLL

Installers—InstallShield

- Problems with Uninstalling
  - InstallShield versions that can call a function from a DLL:
    - Developer 7
    - InstallShield Professional 6.3
  - Other versions require hocus-pocus:
    - Having special entry points in an installed conduit
    - Calling uninstaller with a special flag

Installers—InstallShield

- Two basic approaches to using DLLs:
  - Call Conduit Manager, Install Aide, and HotSync DLLs directly
    - You do this with the InstallShield scripting language
  - Call an intermediary DLL that itself calls the three DLLs



Installers—Installer Vise

- Capabilities:
  - Can call a DLL at install time, and at uninstall time
    - Function in DLL must have specific signature
- Demo limitations:
  - You can’t distribute installers made with it
  - Installers timeout after 3 days



Installers—Pilot Catapult

- Designed specifically for installing Palm OS applications and conduits
- Low-cost commercial version
- Free version
  - No support
  - Fully functional
  - Updated less often than the commercial version
  - Includes notice in installer



Installers—Pilot Catapult

- Doesn’t use Install Aide or Conduit Manager
  - May not work if Conduit Manager changes



Agenda Slide

- Installing and Registering
- DLLs
- Shim DLL
- Commercial Installers
- Q&A



PalmSource®

Conference and Expo